

Problem A. AtCoDeer and Paint Cans

Time Limit 2000 ms

Problem Statement

AtCoDeer the deer recently bought three paint cans. The color of the one he bought two days ago is a , the color of the one he bought yesterday is b , and the color of the one he bought today is c . Here, the color of each paint can is represented by an integer between 1 and 100, inclusive.

Since he is forgetful, he might have bought more than one paint can in the same color. Count the number of different kinds of colors of these paint cans and tell him.

Constraints

- $1 \leq a, b, c \leq 100$

Input

The input is given from Standard Input in the following format:

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 $a$   $b$   $c$ 
```

Output

Print the number of different kinds of colors of the paint cans.

Sample 1

Input	Output
3 1 4	3

Three different colors: 1, 3, and 4.

Sample 2

Input	Output
3 3 33	2

Two different colors: 3 and 33.

Problem B. Grouping

Time Limit 2000 ms

Problem Statement

Based on some criterion, Snuke divided the integers from 1 through 12 into three groups as shown in the figure below. Given two integers x and y ($1 \leq x < y \leq 12$), determine whether they belong to the same group.



Constraints

- x and y are integers.
- $1 \leq x < y \leq 12$

Input

Input is given from Standard Input in the following format:

x y

Output

If x and y belong to the same group, print **Yes**; otherwise, print **No**.

Sample 1

Input	Output
1 3	Yes

Sample 2

Input	Output
2 4	No

Problem C. RGB Cards

Time Limit 2000 ms

Problem Statement

AtCoDeer has three cards, one red, one green and one blue.

An integer between 1 and 9 (inclusive) is written on each card: r on the red card, g on the green card and b on the blue card.

We will arrange the cards in the order red, green and blue from left to right, and read them as a three-digit integer.

Is this integer a multiple of 4?

Constraints

- $1 \leq r, g, b \leq 9$

Input

Input is given from Standard Input in the following format:

r g b

Output

If the three-digit integer is a multiple of 4, print **YES** (case-sensitive); otherwise, print **NO**.

Sample 1

Input	Output
4 3 2	YES

432 is a multiple of 4, and thus **YES** should be printed.

Sample 2

Input	Output
2 3 4	NO

234 is not a multiple of 4, and thus **NO** should be printed.

Problem D. Multiple of 2 and N

Time Limit 2000 ms

Problem Statement

You are given a positive integer N . Find the minimum positive integer divisible by both 2 and N .

Constraints

- $1 \leq N \leq 10^9$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

N

Output

Print the minimum positive integer divisible by both 2 and N .

Sample 1

Input	Output
3	6

6 is divisible by both 2 and 3. Also, there is no positive integer less than 6 that is divisible by both 2 and 3. Thus, the answer is 6.

Sample 2

Input	Output
10	10

Sample 3

Input	Output
999999999	1999999998

Problem E. Watermelon

Time Limit 1000 ms

Mem Limit 65536 kB

Input File stdin

Output File stdout

One hot summer day Pete and his friend Billy decided to buy a watermelon. They chose the biggest and the ripest one, in their opinion. After that the watermelon was weighed, and the scales showed w kilos. They rushed home, dying of thirst, and decided to divide the berry, however they faced a hard problem.

Pete and Billy are great fans of even numbers, that's why they want to divide the watermelon in such a way that each of the two parts weighs even number of kilos, at the same time it is not obligatory that the parts are equal. The boys are extremely tired and want to start their meal as soon as possible, that's why you should help them and find out, if they can divide the watermelon in the way they want. For sure, each of them should get a part of positive weight.

Input

The first (and the only) input line contains integer number w ($1 \leq w \leq 100$) — the weight of the watermelon bought by the boys.

Output

Print YES, if the boys can divide the watermelon into two parts, each of them weighing even number of kilos; and NO in the opposite case.

Examples

Input	Output
8	YES

Note

For example, the boys can divide the watermelon into two parts of 2 and 6 kilos respectively (another variant — two parts of 4 and 4 kilos).

Problem F. Each condition of two

Time Limit 1000 ms

Mem Limit 131072 kB

For the given integer n print "YES", if each of the following conditions holds and "NO" otherwise.

- number n is divisible by 3;
- number n is even and two-digit.

Input

One integer n .

Output

Print "YES" or "NO" depending on conditions.

Examples

Input	Output
12	YES

Input	Output
27	NO

Problem G. Chef Bakes Cake

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

Chef has baked N cakes, each weighing X kilograms. Now, he needs to deliver them.

He has many vehicles, that he will use to transport the cakes. Each vehicle can carry at most Y kilograms of cake. It is guaranteed that $Y \geq X$, so each vehicle can carry at least 1 cake.

How many vehicles will Chef need to transport all the cakes?

Input Format

- The first and only line of each test case contains 3 integers - N , X and Y .

Output Format

Output on a new line the minimum number of vehicles Chef will need.

Constraints

- $1 \leq N \leq 10$
- $1 \leq X \leq Y \leq 10$

Sample 1

Input	Output
6 4 6	6

Each vehicle can only fit 1 cake, and thus Chef will need 6 vehicles for 6 cakes.

Sample 2

Input	Output
4 1 3	2

Chef can put 2 cakes in each of the 2 vehicles.

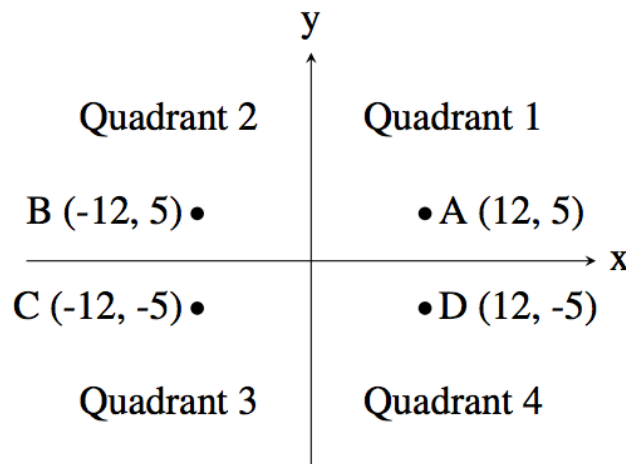
Problem H. Quadrant Selection

Time Limit 1000 ms

Mem Limit 1048576 kB

OS Linux

A common problem in mathematics is to determine which quadrant a given point lies in. There are four quadrants, numbered from 1 to 4, as shown in the diagram below:



For example, the point A , which is at coordinates $(12, 5)$ lies in quadrant 1 since both its x and y values are positive, and point B lies in quadrant 2 since its x value is negative and its y value is positive.

Your job is to take a point and determine the quadrant it is in. You can assume that neither of the two coordinates will be 0.

Input

The first line of input contains the integer x ($-1000 \leq x \leq 1000$; $x \neq 0$). The second line of input contains the integer y ($-1000 \leq y \leq 1000$; $y \neq 0$).

Output

Output the quadrant number (1, 2, 3 or 4) for the point (x, y) .

Sample 1

Input	Output
10 6	1

Sample 2

Input	Output
9 -13	4

Problem I. Olympics 2024

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

Athletes of Chefland have won G gold medals, S silver medals, and B bronze medals in the Paris Olympics.

Chef will be happy only if the Chefland team wins 5 medals of each type. Find the total number of additional medals the team needs to win to make Chef happy.

Input Format

- The first and only line of input will contain three space-separated integers G , S , and B — the number of gold, silver, and bronze medals, respectively.

Output Format

For each test case, output on a new line, the total number of additional medals the team needs to win to make Chef happy.

Constraints

- $1 \leq G, S, B \leq 5$

Sample 1

Input	Output
5 1 4	5

The team already has 5 gold medals. They need 4 silver medals and 1 bronze medal. Thus, a total of 5 medals are needed to make Chef happy.

Sample 2

Input	Output
2 1 1	11

The team needs 3 gold, 4 silver, and 4 bronze medals. Thus, they need a total of 11 medals to make Chef happy.

Problem J. Payment Scheme

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

Chef picked out several items in a shop, and now has to pay for them.

The shop offers Chef two payment schemes:

1. Pay 100 coins immediately, and then pay X coins **every week** for the next **four** weeks.
2. Pay 300 coins immediately, with no future payments required.

Find the minimum possible number of coins Chef needs to pay if he chooses the payment scheme appropriately.

Input Format

- The input contains a single integer X — the required weekly payment for the first payment scheme.

Output Format

Output a single integer: the minimum possible number of coins Chef needs to pay.

Constraints

- $1 \leq X \leq 100$

Sample 1

Input	Output
8	132

In this case, it's ideal to choose the first payment scheme: pay 100 coins immediately, and then pay $X = 8$ coins each week for the next four weeks. The total cost comes out to $100 + 4 \cdot 8 = 132$, which is better than paying 300 via the second option.

Sample 2

Input	Output
77	300

In this case, since $X = 77$ it's better to choose the second scheme and pay 300 coins immediately. The first scheme would lead to a total of $100 + 4 \cdot 77 = 408$ coins which is more expensive.

Sample 3

Input	Output
39	256

Problem K. Cake Making

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

There are exactly 100 colours in the world, numbered 1 to 100.

Chef is making a 2-layered cake, and all that is left is deciding the colours of the 2 layers.

Chef has decided that we can choose any of the colours $1, 2, \dots, A$ for the first layer, and any of the colours $1, 2, \dots, B$ for the second layer.

However, there is an **extra constraint**. To encourage diversity, the first and the second layer should not have the same colour.

How many different cakes are possible while following the above rules? 2 cakes are said to be different when either the first layer or the second layer has a different colour.

Input Format

- The first and only line of input contains 2 integers A and B .

Output Format

Output the total number of possible cakes.

Constraints

- $1 \leq A, B \leq 100$

Sample 1

Input	Output
2 3	4

The following cakes are possible: (The first number represents the colour of layer 1 and the second number the colour of layer 2)

- (1, 2)
- (1, 3)
- (2, 1)
- (2, 3)

Sample 2

Input	Output
1 1	0

There are no possible cakes because we must make sure layer 1 and layer 2 do not have the same colour.

Sample 3

Input	Output
100 5	495

Problem L. Five-digit in order

Time Limit 1000 ms

Mem Limit 131072 kB

Five-digit number is given. Determine whether all its digits are in ascending order, starting with the leftmost one. For example, in the number 15689 all digits are arranged in ascending order.

Input

One five-digit natural number n .

Output

Print "YES", if all digits in number n are arranged in ascending order, and "NO" otherwise.

Examples

Input	Output
15689	YES

Input	Output
12321	NO

Problem M. Exactly one condition out of two

Time Limit 1000 ms

Mem Limit 131072 kB

For a given integer n , print "YES" if exactly one of the following conditions is satisfied, and "NO" otherwise:

- n is an even number;
- n is a negative number and divisible by three.

Input

One single integer n .

Output

Print "YES" or "NO" depending on whether the specified conditions are satisfied.

Examples

Input	Output
22	YES

Input	Output
7	NO

Problem N. Leap Years

Time Limit 1000 ms

Mem Limit 524288 kB

In the Gregorian calendar, certain years have 366 days instead of 365. In such years, the month of February is extended to have 29 days (instead of 28 days). These years are known as leap years.

A leap year occurs when the year is a multiple of 4 but not a multiple of 400.

Given a year, determine if the year is a leap year.

Input

The input will contain a one integer Y ($0 < Y < 9999$).

Output

Print **Yes** if the year is a leap year, otherwise **No**.

Sample

Input	Output
2004	Yes

This problem uses an incorrect definition of leap years. [Another problem](#) exists in our archive that uses the correct definition.

Problem O. Clock Math

Time Limit 1000 ms

Mem Limit 524288 kB

Given a time (hours as H and minutes as M), determine the smaller angle between the two hands of a clock showing the time and print it.



Input

The input will contain two integers: H ($0 \leq H < 12$) and M ($0 \leq M < 60$).

Output

Print the angle in degrees (positive and accurate to 10^{-4}).

Example

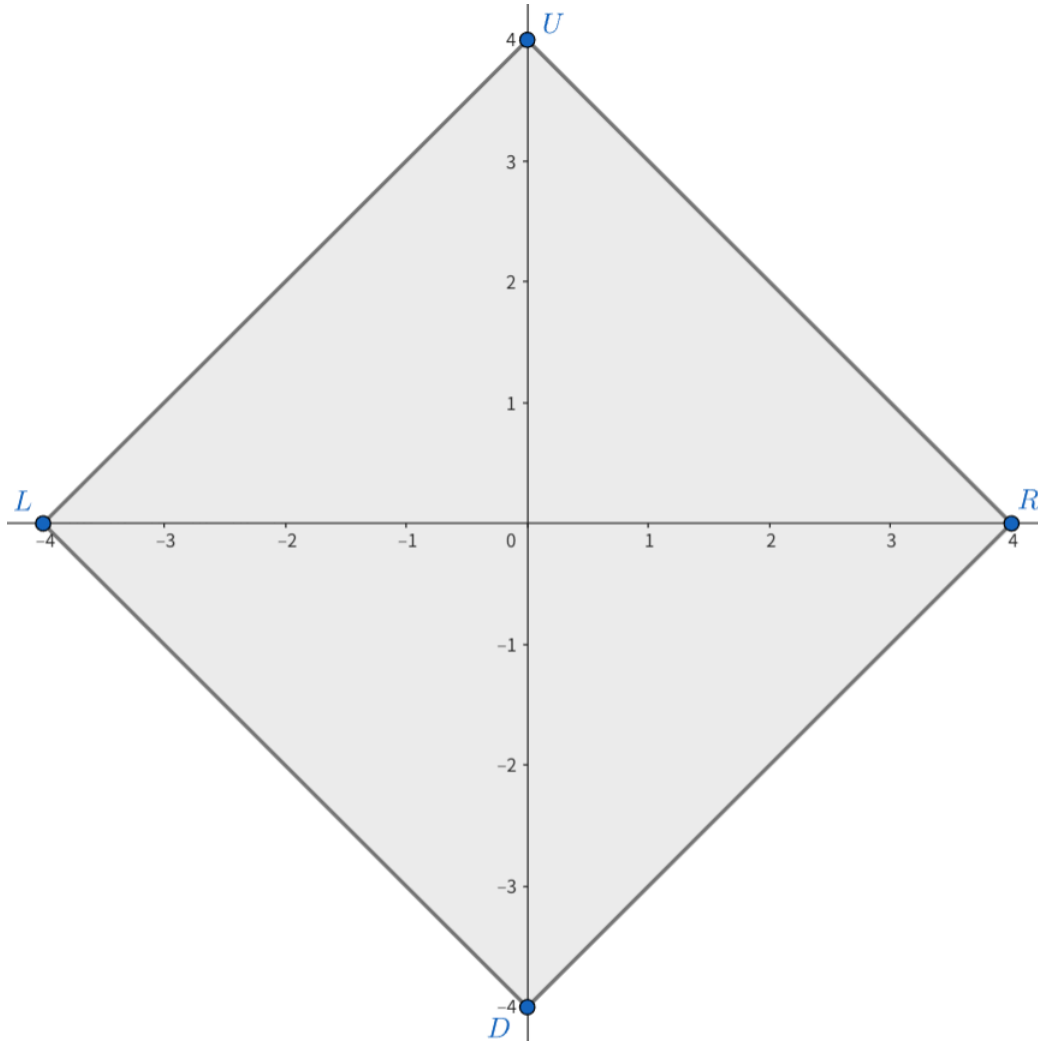
Input	Output
10 15	142.5000000

Problem P. Draw a Square

Time Limit 1000 ms

Mem Limit 262144 kB

The pink soldiers have given you 4 **distinct points** on the plane. The 4 points' coordinates are $(-l, 0)$, $(r, 0)$, $(0, -d)$, $(0, u)$ correspondingly, where l, r, d, u are positive integers.



In the diagram, a square is drawn by connecting the four points L, R, D, U .

Please determine if it is possible to draw a square* with the given points as its vertices.

*A *square* is defined as a polygon consisting of 4 vertices, of which all sides have equal length and all inner angles are equal. No two edges of the polygon may intersect each other.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains four integers l, r, d, u ($1 \leq l, r, d, u \leq 10$).

Output

For each test case, if you can draw a square using the four points, output "Yes". Otherwise, output "No".

You can output the answer in any case. For example, the strings "yEs", "yes", and "YES" will also be recognized as positive responses.

Examples

Input	Output
2 2 2 2 2 1 2 3 4	Yes No

Note

On the first test case, the four given points form a square, so the answer is "Yes".

On the second test case, the four given points do not form a square, so the answer is "No".

Problem Q. Juice

Time Limit 1000 ms

Mem Limit 131072 kB

In building C of ADA University, one prepares a very tasty and healthy multivitamin juice. One dose of juice requires a apples, b pears, and c oranges. However, there is a catch — you must bring the fruits for juice with you!

Huseyn likes multivitamin juice very much. He currently has x apples, y pears, and z oranges. What is the maximum number of juice doses he can order?

Input

The first line contains three integers a , b , and c . The second line contains three integers x , y , and z . All numbers are non-negative integers not greater than 10^9 .

Output

Print the largest number of juice doses that Huseyn can order.

Examples

Input	Output
1 2 6 3 8 15	2

Problem R. Sakurako's Exam

Time Limit 1000 ms

Mem Limit 262144 kB

Today, Sakurako has a math exam. The teacher gave the array, consisting of a ones and b twos.

In an array, Sakurako **must** place either a '+' or a '-' in front of each element so that the sum of all elements in the array equals 0.

Sakurako is not sure if it is possible to solve this problem, so determine whether there is a way to assign signs such that the sum of all elements in the array equals 0.

Input

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

The only line of each test case contains two integers a and b ($0 \leq a, b < 10$) — the number of '1's and the number of '2's in the array.

Output

For each test case, output "Yes" if you can make the sum of the entire array equal to 0, and "No" otherwise.

You can output each letter in any case (lowercase or uppercase). For example, the strings "yEs", "yes", "Yes", and "YES" will be accepted as a positive answer.

Examples

Input	Output
5	NO
0 1	NO
0 3	YES
2 0	YES
2 3	NO
3 1	

Note

1. $a = 0, b = 1$: This means the array is $[2]$ — it is impossible to add the signs '+' or '-' to get 0 as a result;
2. $a = 0, b = 3$: This means the array is $[2, 2, 2]$ — it is impossible to add the signs '+' or '-' to get 0 as a result;
3. $a = 2, b = 0$: This means the array is $[1, 1]$ — it is possible to add the signs '+' or '-' to get 0 as a result ($+1 - 1 = 0$);
4. $a = 2, b = 3$: This means the array is $[1, 1, 2, 2, 2]$ — it is possible to add the signs '+' or '-' to get 0 as a result ($+1 + 1 - 2 - 2 + 2 = 0$);

Problem S. Distanced Coloring

Time Limit 1000 ms

Mem Limit 262144 kB

You received an $n \times m$ grid from a mysterious source. The source also gave you a magic positive integer constant k .

The source told you to color the grid with some colors, satisfying the following condition:

- If $(x_1, y_1), (x_2, y_2)$ are two distinct cells with the same color, then $\max(|x_1 - x_2|, |y_1 - y_2|) \geq k$.

You don't like using too many colors. Please find the minimum number of colors needed to color the grid.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 1000$). The description of the test cases follows.

The only line of each test case consists of three positive integers n, m, k ($1 \leq n, m, k \leq 10^4$) — the dimensions of the grid and the magic constant.

Output

For each test case, print a single integer — the minimum number of colors needed to color the grid.

Examples

Input	Output
6	4
3 3 2	5
5 1 10000	12
7 3 4	6
3 2 7	36
8 9 6	8
2 5 4	

Note

In the first test case, one of the optimal constructions is:

2	1	2
3	4	3
1	2	1

In the second test case, the color of all cells must be pairwise different, so the answer is 5.

Problem T. Adjacent Digit Sums

Time Limit 1000 ms

Mem Limit 262144 kB

You are given two numbers x, y . You need to determine if there exists an integer n such that $S(n) = x, S(n + 1) = y$.

Here, $S(a)$ denotes the sum of the digits of the number a in the decimal numeral system.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 500$). The description of the test cases follows.

The first line of each test case contains two integers x, y ($1 \leq x \leq 1000, 1 \leq y \leq 1000$).

Output

For each test case, print "NO" if a suitable n does not exist. Otherwise, output "YES".

You can output each letter in any case (for example, "YES", "Yes", "yes", "yEs", "yEs" will be recognized as a positive answer).

Examples

Input	Output
7	Yes
1 2	No
77 77	No
997 999	Yes
999 1	No
1000 1	No
1 11	Yes
18 1	

Note

In the first test case, for example, $n = 100$ works. $S(100) = 1, S(101) = 2$.

In the second test case, it can be shown that $S(n) \neq S(n + 1)$ for all n ; therefore, the answer is No.

In the fourth test case, $n = 10^{111} - 1$ works, which is a number consisting of 111 digits of 9.

Problem U. Buns

Time Limit 1000 ms

Mem Limit 131072 kB

Huseyn likes buns very much, which are sold at ADA University in building C. It is known that

- you can buy one bun for a qəpik;
- you can buy three buns for b qəpik;

Huseyn wants to buy exactly n buns. What is the minimum number of qəpik he should spend?

Input

Three positive integers a , b , and n , each not greater than 10^9 .

Output

Print the minimum number of qəpik that Huseyn should spend to buy exactly n buns.

Examples

Input	Output
2 5 10	17

Problem V. Even and odd numbers

Time Limit 1000 ms

Mem Limit 131072 kB

Three integers a, b, c are given. Determine if there is at least one even number and at least one odd number among them.

Input

Three integers a, b, c , each with absolute value not exceeding 10000 (numbers can be negative).

Output

Print **YES** if there is at least one even number and at least one odd number among a, b , and c . Otherwise, print **NO**.

Examples

Input	Output
3 4 5	YES

Input	Output
7 7 7	NO

Problem W. New World, New Me, New Array

Time Limit 1000 ms

Mem Limit 262144 kB

Natsume Akito has just woken up in a new world and immediately received his first quest! The system provided him with an array a of n zeros, an integer k , and an integer p .

In one operation, Akito chooses two integers i and x such that $1 \leq i \leq n$ and $-p \leq x \leq p$, and performs the assignment $a_i = x$.

Akito is still not fully accustomed to controlling his new body, so help him calculate the minimum number of operations required to make the sum of all elements in the array equal to k , or tell him that it is impossible.

Input

The first line of input contains one integer t ($1 \leq t \leq 1000$) — the number of test cases.

The only line of each test case contains three integers n, k, p ($1 \leq n \leq 50$, $-2500 \leq k \leq 2500$, $1 \leq p \leq 50$) — the length of the array, the required sum, and the boundary of the segment from which numbers can be replaced.

Output

For each test case, output the minimum number of operations to achieve the final sum k in the array, or -1 if it is impossible to achieve the sum k .

Examples

Input	Output
8	10
21 100 10	-1
9 -420 42	4
5 -7 2	6
13 37 7	0
10 0 49	-1
1 10 9	1
7 -7 7	-1
20 31 1	

Note

In the fifth example, the sum of the array is initially zero, so no operations are needed.

In the sixth example, the maximum sum in the array that we can achieve is 9 (by assigning the number 9 to the single element), so the sum 10 cannot be obtained by any operations.

In the seventh example, only one operation $a_3 = -7$ is needed.

Problem X. Quadratic equation

Time Limit 1000 ms

Mem Limit 131072 kB

Solve the quadratic equation $a \cdot x^2 + b \cdot x + c = 0$, where $a \neq 0$.

Input

The input consists of one line containing three integers — the coefficients of the quadratic equation a , b , and c . The absolute values of all coefficients do not exceed 100.

Output

Print on a single line:

- **No roots**, if the equation has no roots;
- **One root**: followed by the root, separated by a space, if the equation has one root;
- **Two roots**: followed by the smaller root and the larger root, separated by a space, if the equation has two roots.

It is guaranteed that if solutions exist, all roots are integers.

Examples

Input	Output
1 -5 6	Two roots: 2 3

Input	Output
1 2 10	No roots

Problem Y. Wardrobe

Time Limit 1000 ms

Mem Limit 131072 kB

A wardrobe has dimensions $a \times b \times c$. Is it possible to carry it through a doorway of size $x \times y$? The wardrobe can pass through the doorway if there exist two of its dimensions such that, after possibly rotating the wardrobe, both are less than or equal to the corresponding dimensions of the doorway.

Input

Integer numbers a, b, c, x, y ($1 \leq a, b, c, x, y \leq 100$).

Output

Output **YES** if it is possible to carry the wardrobe through the doorway; otherwise, output **NO**.

Examples

Input	Output
4 5 6 10 20	YES

Input	Output
6 5 4 4 5	YES

Problem Z. Four-digit palindrom

Time Limit 1000 ms

Mem Limit 131072 kB

A four-digit natural number is given. Determine whether it is a palindrome. A natural number is a palindrome if it reads the same from left to right and from right to left. For example, the number 2332 is a palindrome.

Input

One four-digit natural number n .

Output

Print "YES", if the number n is a palindrome, and "NO" otherwise.

Examples

Input	Output
2332	YES

Input	Output
5446	NO